MILITARY SPECIFICATION

ARMOR, BODY, FRAGMENTATION, PROTECTIVE: FOOTWEAR

1. SCOPE

1.1 Scope. This specification covers requirements for fragmentation protective body armor footwear, which consists of a right and left footwear.

1.2 Classification. The footwear body armor shall be of one type in the following sizes as specified (see 6.2):

Schedule of sizes (to fit combat boot size (range)

Regular - 6 narrow through 8 wide
Large - 9 narrow through 11 wide

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

C-F-206	- Felt, Sheet, Wool, Pressed.
V-T-295	- Thread, Nylon.
QQ-S-763	- Steel Bars, Shapes, And Forgings, Corrosion-Resisting.
QQ-S-764	- Steel Bar, Corrosion-Resisting, Free Machining.
QQ-S-766	- Steel Plates, Sheets, And Strip-Corrosion-Resisting.
PPP-B-585	- Boxes, Wood Wirebound.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed And Lock-Corner.
PPP-B-636	- Box, Fiberboard
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-P-116	- Preservation, Methods Of.
MIL-R-3065	- Rubber, Fabricated Parts.
MIL-A-8625	- Anodic Coatings,For Aluminum And Aluminum Alloys.
MIL-L-10867	- Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored.
MIL-C-13924	- Coating, Oxide/Black,For Ferrous Metals.
MIL-E-19933	- Electrodes And Rods-Welding, Bare, Chromium-Nickel Steels.
MIL-C-20696	- Cloth, Coated, Nylon, Waterproof.
MIL-F-21840	- Fastener Tape, Hook And Pile, Nylon.

STANDARDS

FEDERAL

FED-STD-595	- Colors.
FED-STD-751	- Stitches, Seams, And Stitchings.

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MIL-STD-105	- Sampling Procedures And Tables For Inspection By Attributes.
MIL-STD-129	- Marking For Shipment And Storage.
MIL-STD-401	- Sandwich Constructions and Core Materials; General Test Methods.
MIL-STD-417	- Rubber Compositions, Vulcanized General Purpose, Solid (Symbols And Tests).
MS15795	- Washer, Flat-Metal, Round, General Purpose.
MS16535	- Rivet, Tubular Oval Head.
MS20435	- Rivet, Round Head, Steel, Monel And Copper.
MS171401 Thru MS171900	- Pin-Spring, Corrosion Resistant Steel.

DRAWINGS

MARINE CORPS

P-235-1	- Armor, Body, Fragmentation, Protective: Footwear (Assembly).
P-235-2	- Armor, Body, Fragmentation, Protective: Footwear (Hull).
P-235-3	- Armor, Body, Fragmentation, Protective: Footwear (Hull Section).
P-235-4	- Armor, Body, Fragmentation, Protective: Footwear (Plate Assembly).
P-235-5	- Armor, Body, Fragmentation, Protective: Footwear (Strap Assembly Inner Front And Inner Rear).
P-235-6	- Armor, Body, Fragmentation, Protective: Footwear (Roll Clamp).
P-235-7	- Armor, Body, Fragmentation, Protective: Footwear (Pivot-Upright).
P-235-8	- Armor, Body, Fragmentation, Protective: Footwear (Brace Assembly).
P-235-9	- Armor, Body, Fragmentation, Protective: Footwear (Strap Assembly Outer Front).
P-235-10	- Armor, Body, Fragmentation, Protective: Footwear (Loop Assembly).
P-235-11	- Armor, Body, Fragmentation, Protective: Footwear (Hull Insert Plastic Form).
P-235-12	- Armor, Body, Fragmentation, Protective: Footwear (Tread Detail).
P-235-13	- Armor, Body, Fragmentation, Protective: Footwear (Loop-Large).
P-235-14	- Armor, Body, Fragmentation, Protective: Footwear (Pivot Brace).
P-235-15	- Armor, Body, Fragmentation, Protective: Footwear (Strap Assembly Brace).
P-235-16	- Armor, Body, Fragmentation, Protective: Footwear (Rivet-Tubular).
P-235-17	- Armor, Body, Fragmentation, Protective: Footwear (Brace).
P-235-18	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Upper Outer Stamping).
P-235-19	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Upper Inner Stamping).
P-235-20	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Lower Outer-Large Stamping).
P-235-21	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Lower Inner-Large Stamping).
P-235-22	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Lower Outer-Regular Stamping).
P-235-23	- Armor, Body, Fragmentation, Protective: Footwear (Upright-Lower Inner-Regular Stamping).

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(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS

D735-61-T - Method of test for Elastomer Compounds for Automotive Application (tentative); Test Methods - D1667, D256, D730

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

NATIONAL AIRCRAFT STANDARDS COMMITTEE

NAS-508 - Rivet-Universal Head Monel

(Application for copies should be addressed to the Aircraft Industries Association of America, Inc., 610 Shoreham Building, Washington, D. C. 20005.)

3. REQUIREMENTS

3.1 Samples. Samples are furnished solely for guidance and information to the supplier (see 6.3). Variation from the specification may appear in the sample, in which case the specification shall govern.

3.2 Preproduction sample approval. Unless otherwise specified (see 6.2), before production is commenced, a sample of the finished armored body footwear shall be submitted, or made available, to the contracting officer or his authorized representative for approval. After approval of the preproduction sample, the approved sample may be retained by the supplier for inspection comparison purposes throughout production. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all provisions of this specification. The preproduction sample shall be manufactured by the supplier in the same facilities to be used for the manufacture of the production item, and shall be representative of the construction and quality of workmanship to be expected of the subsequent production item.

3.3 Material. Materials used in the manufacture of the armored, body, footwear shall conform in all respects to the specifications cited herein and Drawings P-235-1 through P-235-23.

3.3.1 Felt padding. The padding for the brace shall be felt, 1/8 inch thick, conforming to type I, classification number 9 R, class 4 of C-F-206.

3.3.2 Leather. The cover for the brace shall be leather, black shade, conforming to treatment B of MIL-L-10867.

3.3.3 Washer. The washers shall be fabricated of corrosion-resisting steel conforming to class 304, condition A, low magnetic permeability, of QQ-S-766. The dimensions for the outside washers of the brace shall conform to MS15795-808 and the dimensions for the outside washers of the uprights shall conform to

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MS15795-808. The dimensions for the inside washers of the brace shall conform to MS15795-810. The protective coating, treatment and magnetic permeability specified in MS15795 shall not apply.

3.3.4 Nylon fastener tape. The tape shall be nylon, hook and pile tape, black shade, conforming to type I, 1-inch in width of MIL-F-21840.

3.3.5 Thread, nylon. The nylon thread shall conform to type II, class 1, size E, 3-ply of V-T-295.

3.3.5.1 Color and colorfastness. Black thread shall be used for seaming and stitching the tape. The dyed thread shall show fastness to laundering and crocking equal to or better than the standard sample. When no standard sample is available, the dyed thread shall show good fastness to laundering and crocking.

3.3.6 Loops. The loops large and small shall be made from corrosion-resisting steel bars, conforming to class 304, condition A, cold drawn, low magnetic permeability of QQ-S-763.

3.3.7 Brace. The brace shall be made from corrosion-resisting steel sheets, conforming to class 304, condition A, No. 2 B finish, low magnetic permeability of QQ-S-766.

3.3.8 Pivot. The pivot for the brace and uprights shall be fabricated from round bars, conforming to class 305, condition A, cold drawn, low magnetic permeability of QQ-S-763; or Carpenter No. 10 corrosion-resisting steel or equal, upon approval by the contracting officer.

3.3.9 Rivets. The rivets for the pivot brace and uprights shall conform to part No. 508M8-10 of NAS-508, except material (see 3.3.8).

3.3.10 Uprights. The upper and lower uprights for supporting the brace shall be made from corrosion-resisting steel flat bars, conforming to class 304, condition A, cold drawn, and pickled, low magnetic permeability of QQ-S-763.

3.3.11 Rivet, tubular. The rivet for the brace shall be made from round bars, conforming to class 305, condition A, low magnetic permeability of QQ-S-763; or Carpenter No. 10 or equal, corrosion-resisting steel upon approval by the contracting officer. The dimension for the rivet shall conform to Drawing No. P-235-16.

3.3.12 Roll clamp. The roll clamp shall be made from steel bars conforming to class 304, condition A, annealed and centerless ground of QQ-S-763. Alternately, the roll clamp may be made from steel bars conforming to class 303Ma, condition A, annealed and centerless ground of QQ-S-764. Either material shall meet the requirements for low magnetic permeability of QQ-S-763.

3.3.13 Plate. The plate shall be made from corrosion-resisting steel sheets, conforming to class 304, condition A, No. 1 finish, low magnetic permeability of QQ-S-766.

3.3.14 Pin, spring. The pin spring shall conform to MS171654, except the pin spring shall be made from corrosion-resisting steel sheets, conforming to class 304, condition A, No. 1 finish, low magnetic permeability of QQ-S-766.

3.3.15 Rivet, round head. The rivet for attaching the roll clamp and plate together shall conform to MS20435-F3-10.

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3.3.16 Rivet, tubular oval head. The rivet for attaching the inner rear fastening tape together shall conform to aluminum alloy of MS16535-338.

3.3.17 Hull. The hull (see 6.4) shall be made from corrosion-resisting steel sheets, conforming to class 304, condition A, No. 1 finish, low magnetic permeability of QQ-S-766.

3.3.18 Plastic foam filler. The plastic foam filler for the hull shall be polystyrene and shall conform to the following requirements (see Section 4, Table IV for required test).

Density - 4.5 to 5.5 lbs/ft.^{3}
Impact strength - 0.20 in. (min.)
Flexural strength - 120 lbs/sq. inch (min.)

3.3.19 Rubber tread. The rubber tread shall meet the requirements of type R, class RS, grade RS and suffix letter RS 720 GJK_2L of MIL-STD-417 and MIL-R-3065. The color shall be black conforming to color No. 37038 of FED-STD-595.

3.3.20 Cloth, coated, nylon. The cloth, nylon, coated shall conform to type II, class 1 of MIL-C-20696. The color shall be black on both sides and match black No. 37038 of FED-STD-595.

3.3.21 Rod-welding. Bare welding rod for welding type 304 stainless steel material shall conform to type MIL-308 of MIL-E-19933.

3.3.22 Adhesive. The felt, leather, rubber, metal, combinations shall be cemented with Bostik adhesive No. 1096, or equal; equivalent shall be approved by the contracting officer.

3.3.23 Cement. The neoprene impregnated cloth, polystyrene foam, combinations shall be cemented using Bostik adhesive No. 1015, or equal; equivalent shall be approved by the contracting officer.

3.3.24 Primer. All metallic surfaces to which adhesives or neoprene coatings are to be applied shall be primed prior to application of applicable adhesives, second coatings with Bostik adhesive No. 1007, or equal, primer or Gaco N-15, or equal, neoprene primer; equivalents shall be approved by the contracting officer.

3.3.25 Black oxide coating. All exposed corrosion-resisting steel surfaces shall be black oxide coated. The coating, shall conform to class 2 or 3 of MIL-C-13924. Coatings shall be applied after surface preparation (see 3.4.4.1).

3.3.26 Anodize. All exposed aluminum surfaces shall be treated by anodizing. The anodic coating shall conform to type I or II, class 2, dyed black, of MIL-A-8625.

3.3.27 Size marking. The size shall be die stamped on the outside of hull at upper heel section.

3.3.28 Stitches, seams, and stitching. The stitch, seam and stitching type shall conform to FED-STD-751, (see 3.4.3.3.1).

3.4 Construction. Construction of the armored footwear shall conform in all respects to the design, details, dimensions and materials as shown on Drawings P-235-1 through P-235-23 and as specified herein. The armored footwear assembly is made up of three basic functional areas: the hull, brace, and fastening tape.

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3.4.1 Hull.

3.4.1.1 Hull assembly. The hull assembly shall consist of the following:

(a) Hull. Fabricated of corrosion-resisting steel shall be as specified in 3.3.17. All interior and exterior surfaces of the hull shall be coated with synthetic rubber, or neoprene or equal as approved by the contracting officer.

(b) Hull filler. Fabricated of polystyrene plastic foam shall conform to 3.3.18, completely filling the hull as per guide sample (see 3.1).

(c) Coated cloth. Fabricated of coated nylon cloth shall conform to 3.3.20.

(d) Rubber tread. Fabricated of synthetic rubber shall conform to 3.3.19.

3.4.1.2 Hull fabrication. The vertex angle of the wedge shape shall be 60° included from mid-length forward, and taper toward the back to 47° at the beginning of the heel section as per guide sample (see 3.1), be made of conical configuration with an angle of 7° at the back measured in a plane normal to the above cross section angles. The vertex radius shall be 0.080 inch maximum, and the height from the vertex radius shall be 4 inches minimum to 4-5/8 inches maximum for regular and large sizes inclusive.

3.4.1.3 Contour of hull. The hull deck contour shall be sized from regular to large size (see Drawing P-235-2) to accommodate the size range of combat boots shown in 1.2. The hull deck contour shall blend smoothly to the vertex as shown on the guide sample (see 3.1), to the angles given, and within the height limitations specified in 3.4.1.2. The contour of hull shall finish smooth.

3.4.1.4 Insert plastic foam. The hull fabrication shall be filled with expended polystyrene foam specified in 3.3.18, forming an insert for shoe support and detonation impact absorption. Filler insert shall be formed at the deck line to accommodate the horizontal boot sole contour with a step recess at the rear below the hull sides to fit and support the boot heel. The filler shall be cemented to the hull with Bostik adhesive No. 1015, or equal; equivalent shall be approved by the contracting officer. The filler shall be covered with coated nylon cloth, and shall conform to 3.3.20, to provide protection for the foam filler during service. The coated nylon cloth shall be cemented to the deck of the plastic foam insert with cement and shall conform to 3.3.24.

3.4.1.5 Synthetic rubber tread. The rubber tread shall conform to 3.3.19 and shall be vulcanized to the bottom of the synthetic rubber coated corrosion-resisting steel hull, using the adhesive specified in 3.3.22.

3.4.1.6 Bonding. The filler, cover, tread and hull shall be firmly bonded to provide an integral assembly. The adhesives shall be applied to all joining surfaces in an adequate and uniform manner as recommended by the adhesive manufacturer. Assembly periods, clamping or pressing time shall be in accordance with the manufacturer's adhesive instructions.

3.4.2 Brace assembly. The brace assembly shall consist of two sets of uprights, outer and inner, the lower being permanently attached to the hull by welding; the upper being attached to a curved padded ankle support by riveting. The upper and lower uprights shall be joined in the middle also by riveting, all

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riveting being performed so that the joints are free to pivot, with all rivet holes in the same plane, to accommodate motion of the ankle relative to the foot. All burrs and sharp edges shall be removed from metallic components prior to surface coating (as applicable) and assembly.

3.4.2.1 Padding. The brace shall be padded on the inside with felt conforming to 3.3.1, covered entirely with leather conforming to 3.3.2, and located approximately 7-5/8 inches from the hull sole plane to a pivot at approximately the brace vertical center. Prime the brace all over with primer conforming to 3.3.24. The padding is applied to the brace as follows: cut one ply of felt to fit inside of brace and attach felt to brace using adhesive conforming to 3.3.22. Cut the leather (cover) for the outside of brace approximately 2-1/8 inches wide by 7-3/4 inches long. Cut the leather (lining) for the inside of brace approximately 3-1/8 inches wide by 8-3/4 inches long and overlapping the leather approximately 1/2 inch on four sides. Cement the leather in place using adhesive conforming to 3.3.22.

3.4.2.2 Brace uprights. The pivoted axis of the brace assembly shall be approximately 4-1/2 inches above the hull sole plane, and at an angle of approximately 10-1/2° from the normal to the hull center line, the inside pivot being in front of the outside pivot. The inner and outer lower uprights shall be spot welded to the hull in the location shown on Drawing No. P-235-2.

3.4.3. Fastening tape. The assembled armored, body, footwear shall be provided with fastening tape of nylon material conforming to 3.3.4.

3.4.3.1 Hull fastening tape. The hull tapes shall be attached to ears integral with the hull at the toe, and middle of boot at a point approximately over the arch of the foot. Ears shall be provided with pin springs (see 3.3.14) assembled to prevent chafing or cutting of fastening tapes. Toe fastening tapes shall act to hold the front of the armored, body, footwear in contact with the sole of the combat boot, while arch fastening tapes shall act diagonally to hold the rear of the armored, body, footwear against the heel of the boot. The front fastening tape ears shall be constructed so as to provide lateral support for the combat boot. Each toe fastening tape shall also be provided with roll clamps which act at the upper corner of the combat boot sole to provide further clamping action when the toe fastening tapes are fastened.

3.4.3.2 Brace fastening tape. The armored footwear shall be provided with a brace fastening tape located around the top of the brace and acting to hold the brace firmly against the back of the ankle.

3.4.3.3 Fastening tape lengths. The tape shall be cut in the following lengths:

Type of tape	Quantity cut (each)	Lengths inches	Component part location
pile	1	14-1/4 ± 1/4	around top of brace
hook	1	22-1/4 ± 1/4	around top of brace
pile	1	7 ± 1/8	inner front of hull
hook	1	11 ± 1/8	inner front of hull
pile	1	11-1/2 ± 1/8	rear of hull
hook	1	23 ± 1/8	rear of hull
hook	1	6-1/8 ± 1/8	attaching to larger loop and roll clamp - outer

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3.4.3.3.1 Assemble fastening tape. Each fastening tape shall be joined in accordance with the patterns shown on Drawing Nos. P-235-5 and P-235-15. All stitching shall conform to FED-STD-751 and as follows:

(a) For all stitching, type 301, seam type SSa-1, 10-12 stitches per inch, using the thread specified in 3.3.5 for needle and bobbin.

(b) Any of the stitching operations may be performed with an automatic stitching machine provided the requirements for stitch pattern, stitches per inch, size and type of thread are met; and at least three or more tying, overlapping or back stitches are used to secure the ends of stitching.

(c) Thread breaks in stitching shall be over stitched not less than 1 inch at each break.

(d) Stitching shall be backstitched not less than 1/2 inch at the ends.

(e) Thread tension shall be maintained so that there will be no loose stitching and the lock shall be embedded in the materials sewed.

(f) All thread ends shall be trimmed.

3.4.4 Surface coating.

3.4.4.1 Preparation. The inner and outer lower uprights shall be spot welded to the hull prior to surface cleaning. All metallic surfaces shall be prepared prior to surface coating or cementing by abrasive blast cleaning, acid pickling, salt bath descaling, to provide a low reflecting flat matte appearance, or other method approved by the contracting officer. Holes optional.

3.4.4.2 Neoprene coating. All surfaces of the hull and uprights to elevation shown on Drawing No. P-235-1 shall be coated with two coats (rapid succession approximately 10 minutes between coats), of Gaco N-55, or equal, black neoprene coating; equivalent shall be approved by the contracting officer. Thickness of coating shall be 0.005 inch, (see Section 4, Table IV for required test). The neoprene coating shall blend with rubber tread and insert cover to give uniform appearance.

3.4.5 Marking for identification. The marking for identification shall be on one side of the outside of the hull at the upper heel section, and shall consist of the letters "U.S." in characters 1/4-inch high, manufacturer's identification and year of manufacture or year of contract in characters 3/16-inch high. The marking shall be indented (die stamped) prior to surface preparation (see 3.4.4.1) and shall be readily identifiable on completion of application of the synthetic rubber coating.

3.5 Workmanship. The finished armored, footwear shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the levels set by the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Preproduction sample approval. Inspection and testing of preproduction samples shall be made to determine conformance with requirements specified herein.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Component and material inspection. Components and material shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

4.3.1.1 Component testing. The quality assurance provisions for components and materials, except as otherwise required by this specification, shall be in accordance with the applicable specification and standards referenced herein. When there is a conflict in quality assurance provisions, this specification shall govern.

4.3.2 In-process inspection.

4.3.2.1 In-process examination. Examination shall be made during the manufacturing process for conformance to the requirements indicated in Table I. Whenever nonconformance is noted, correction shall be made to affected items and process.

Table I - In-process examination

Requirement	Req Para
Rivets and pivots are specified type.	3.3.15, 3.3.16 and Dwg Nos. P235-16, P235-7, and P235-14
Pin springs are specified type.	3.3.14
Washers are specified type.	3.3.3
No gap at ends of loop wire.	Dwg No. P-235-1
Welding of ends of wire loops for fastening tape.	Dwg No. P-235-6
Metal surface preparation (all metal components).	3.4.4.1
Anodizing aluminum surfaces.	3.3.26
Application of primer on surfaces to which adhesive will be applied.	3.3.24
Application of synthetic rubber coating.	3.4.4.2
Attachment of felt to braces.	3.4.2.1
Overlapping and attachment of leather.	3.4.2.1
Cutting of fastening tape.	3.4.3.3

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Table I - In-process examination (Cont'd)

Requirement	Req Para
Bonding of foam filler.	3.4.1.6
Bonding of nylon cloth cover.	3.3.23, 3.4.1.4 and 3.4.1.6
Bonding of rubber tread.	3.3.22, 3.4.1.5 and 3.4.1.6

4.3.3 Intermediate inspection.

4.3.3.1 Intermediate visual examination of hull (with lower uprights attached). The hull with lower uprights attached , immediately after surface preparation (see 3.4.4.1), shall be examined in accordance with Table II. The sample unit shall be one hull with lower uprights attached . The inspection level for each lot formation shall be level I and the AQL's shall be 1.0 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

Table II - Intermediate examination of hull.

Examine	Defect	Classification Major	Classification Minor
Design and contour	Does not conform to the applicable drawings.	X	
Quality of metal	Broken, fractured, cracked or split.	X	
	Component malformed or bent out of shape.	X	
	Dent in metal.		X
	Burr, fin or sliver.		X
	Machine or tool mark.		X
	Hole or edge not trimmed neatly.		X
	Unspecified hole.	X	
	Surface not clean, e.g., dirt or other foreign matter imbedded.		X
Construction and workmanship	Any part not connected or not joined as specified.	X	
	Component missing.	X	
	Operation omitted or not properly accomplished.	X	

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Examine	Defect	Classification Major	Minor
Welding	Not welded where required.	X	
	Wrong type weld.		X
	Weldment not smooth or not uniform.		X
	Metal deformed at weld.		X
	Metal burned or damaged in welding.	X	
	Weld undercut, overlapped, or contains insufficient leg.		X
	Any weld incomplete, fractured, cracked or not fused.	X	
	Fillet weld not ground to convex contour.		X
	Weld spatter.		X
	Fillet or square weld accomplished on wrong side of component.		X
	One spot weld missing.		X
	More than one spot weld missing.	X	
	Spot weld not in location specified.		X

4.3.3.2 Intermediate visual examination of plastic foam insert. The plastic foam insert after fabrication and immediately prior to installation in the hull shall be examined in accordance with Table III. The sample unit shall be one plastic foam insert. The inspection level for each lot formation shall be level I with AQL's of 1.0 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

Table III - Intermediate examination of plastic foam insert

Examine	Defect	Classification Major	Minor
Design and contour	Does not conform to the applicable drawing.	X	
	Contour distorted.		X
Construction	Not molded in one piece.	X	
Quality of foam	Broken.	X	
	Cracked or split.		X

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Examine	Defect	Classification Major	Minor
	Void, hole, crater, or chip-out on deck or top edge.	X	
	Void, hole, crater or chip-out up to 1/4 inch in longest dimension on other than deck or top edge.		X
	Void, hole, crater or chip-out 1/4 inch or more in longest dimension on other than deck or top edge.	X	
	Porous (presence of numerous visible voids) or contains pockets.		X
	Foreign matter imbedded.		X
	Fish eye, blisters, or evidence of other incomplete fusion.		X

4.3.3.2 Intermediate dimensional examination of parts. A dimensional examination shall be made on the following unassembled parts to determine compliance with this specification and drawings specified herein:

(a) Hull with lower uprights attached, each size.

(b) Fastening tape (cut lengths prior to sewing (see 3.4.3.3).

(c) Leather cover and lining (cut lengths prior to attachment (see 3.4.2.1).

(d) Plastic foam insert, each size.

(e) Washers.

(f) Pivots.

(g) Rivets.

(h) Braces.

(i) Pin springs.

(j) Rubber tread, each size.

The lot shall be all components of one kind. The sample unit shall be one of each kind of component. The inspection level shall be S-3 with an AQL of 2.5, expressed in terms of defects per hundred units.

4.3.3.3 Intermediate testing.

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4.3.3.3.1 Intermediate testing of plastic foam insert. Testing of the molded plastic foam insert shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit shall be four (2 pair) molded plastic foam inserts of each size (regular and large), prior to installing in the hull. The inspection level for each lot formation shall be S-1 and the AQL for material identification , shall be 2.5, expressed in terms of defects per hundred units. The requirement for material identification is applicable to the individual unit. All other requirements, shown in Table IV, are applicable to the lot average.

Table IV - Intermediate testing of plastic foam insert

Characteristic	Req para	Test method	No.Det per indiv. unit	Results reported as
Material identification	3.3.18	Std Comml	1	Pass or fail
Density	3.3.18	ASTM * D1667	2	To nearest 0.1 lb/cu. ft.
Flexural strength	3.3.18	D730 *	2	To nearest 1.0 lbs/sq. in.
Impact strength	3.3.18	D256 *	2	To nearest 0.1 in.lbs/in.

* Method of tests (see ASTM D735-61-T).

(a) Specimens. Finished forms shall be sectioned to produce rind-free specimens appropriate for the performance of the tests. Where production pieces do not provide sufficient material to permit the cutting of the required specimens, sample boards from which these specimens can be cut may be molded or formed by the same procedures and under the same conditions as the production forms. No special treatments shall be used to improve the properties of the specimens when compared with articles molded in commercial production.

(b) Conditioning of specimens. Unless otherwise specified in the contract or order, all tests shall be conducted at a temperature of 23 $\pm$1.1°C (73 $\pm$2°F).

(c) Density. The density of the material shall be determined by dividing the weight of the material by its volume, and shall be expressed in pounds per cubic foot. The volume shall be determined by measuring the volume of water displaced by the material when the material is submerged. Three determinations shall be made.

(d) Flexural strength. Specimens shall be cut from material of each sample unit and shall be 6 by 2 by 1 inch in dimension. The procedure utilized shall be that for simple beam flexure, flatwise loading, of sandwich construction as specified in MIL-STD-401, except that the flexing force shall be applied at a rate of 0.05 inch per minute.

(e) Impact strength (falling ball). Specimens shall be cut from material of each sample unit and be 4 by 4 by 1 inch in dimension. Exterior skin or rind need not be removed for this test. The specimens shall be placed flat upon a solid concrete surface and a steel ball weighing 1.44 pounds, 2.14 inches in diameter, shall be dropped from a height of four feet to the center of the specimen. The ball shall then be removed and after a recovery period of 15 minutes, the maximum depth of the depression measured. At least three specimens from each sample unit shall be so tested.

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4.3.3.3.2 Intermediate testing of coated hull. Testing of the black neoprene coated hull shall be performed in accordance with Table V for the characteristic shown therein. The sample unit shall be one coated hull, with lower uprights attached . The inspection level for each lot formation shall be S-1. The requirements are applicable to the lot average.

Table V - Intermediate testing of coated hull

Characteristic	Req para	Test method	No.Det per indiv. unit	Results reported as
Coating thickness	3.4.4.2	Std Comml	Avg. of 3 determinations	To nearest 0.0001 inch

4.3.4 End item inspection.

4.3.4.1 Examination of the end item. The lot shall consist of all completely fabricated armor, body, fragmentation, footwear (of one size) offered for inspection at one time. The defects found during examination shall be classified in accordance with 4.3.4.1.1 and 4.3.4.1.2. The acceptable quality levels and inspection levels shall be as specified in 4.3.4.1.3.

4.3.4.1.1 Visual examination. The footwear shall be examined for defects in pairing, size, design, material, finish, construction, assembly, workmanship, and marking as indicated in Table VI. The sample unit shall be one completely fabricated footwear and the selection shall be by pairs. The lot size shall be expressed in units of one completely fabricated footwear of one size. Defects of pairing shall be classified as a single defect.

Table VI - End item examination

Examine	Defect	Classification Major	Minor
Pairing and size	Not properly mated, i.e., not right and left of same size.	X	
	Tread improperly matched.		X
	Wrong size.	X	
Design and material	Design of component does not conform to the applicable drawing.	X	
	Wrong material.	X	
	Any metal component which can be attracted by magnet (use magnet and check all parts, including washers and rivets).	X	
Finish of components	Any component not black in color.	X	
	Wrong type coating.	X	
	Area of no finish or no black color.		X
	Black parts not uniform in appearance.		X

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Examine	Defect	Classification Major	Minor
	Discoloration.		X
	Finish not smooth, e.g., contains run, sag, or bead.		X
	Finish not adherent, e.g., blistered, peeled or flaking.	X	
	Foreign matter imbedded in finish.		X
	Rubber coating does not end on lower brace in location shown on drawing.		X
Construction, assembly and workmanship (general)	Component missing.	X	
	Metal part fractured, broken, cracked, or split.	X	
	Part deformed or bent out of shape, seriously affecting serviceability.	X	
	Part deformed or bent out of shape, affecting serviceability but not seriously.		X
	Operation omitted.	X	
	Operation not properly performed, seriously affecting serviceability.	X	
	Operation not properly performed, affecting serviceability but not seriously.		X
	Part misplaced or not in proper alignment.	X	
	Component loose or insecurely fastened.	X	
	Not connected or not joined as specified.	X	
	Part contains rough or sharp edge, burr, fin, metal splinter, or sliver in location that may be injurious to personnel or damage clothing.	X	
	Part contains rough or sharp edge, burr, fin, metal splinter, or sliver not in location that may be injurious to personnel or damage clothing.		X
	Machine or die mark.		X

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Examine	Defect	Classification Major	Minor
Vulcanizing and cementing	Not vulcanized, where required.	X	
	Delamination or area of no bond over 1 inch in length.	X	
	Delamination or area of no bond up to and including 1 inch in length.		X
	Evidence of poor adhesion.	X	
	Air pocket.		X
	Beads.		X
Rubber tread	Void in rubber, sink mark, air cell, or pulled surface over 1 inch in length.	X	
	Void in rubber, sink mark, air cell, or pulled surface up to and including 1 inch in length.		X
	Malformed or out of specified shape.		X
	Loss of tread detail.		X
	Blister or open bubble.		X
	Cut or tear over 1/2-inch deep.	X	
	Cut or tear up to and including 1/2-inch deep.		X
	Surface bruised, cracked, chipped, checked, or otherwise defective.		X
	Area of porosity or sponginess.		X
	Sticky or tacky, as from undercure.		X
	Evidence of deterioration.	X	
	Fin or flash, i.e., not cleanly trimmed.		X
	Configuration of inside of tread does not match outside of hull (at any point).	X	
	Top of tread not chamfered as shown on drawing.		X
Coated cloth	Uncoated area.	X	
	Thinly coated area over 1 square inch (light area or window).	X	

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Examine	Defect	Classification Major	Minor
	Thinly coated area up to and including 1 square inch (light area or window).		X
	Separation of coating from base cloth.	X	
	Blister, lump, bulge or pocket.		X
	Wrinkle or crease on coated surface.		X
	Cut, hole or tear.	X	
	Surface abraded or bruised.		X
	Discoloration or foreign material imbedded.		X
	Surface sticky or tacky.		X
Leather cover and lining	Not attached grain out.	X	
	Grain surface not smooth.		X
	Color not as specified.	X	
	Stain or discoloration.		X
	Boney, boardy, or flanky leather, or contains pronounced veins.		X
	Cut (except where rivet is inserted), hole, tear, or rip.	X	
	Grain damage, scratch, cracked, abraded or wrinkled.		X
	Not turned under neatly, where required.		X
	Not trimmed neatly at edges.		X
	Flap, where rivet is inserted, not cemented.	X	
Fastening tape	Cut, hole or tear.	X	
	Selvage more than 3/32 inch wide.		X
	Spot, stain or discolored.		X
	Broken or missing yarn.		X
	Missing hooks.	X	

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Examine	Defect	Classification Major	Minor
	Placement or attachment of hook and pile tapes not on side specified.	X	
	Tape on brace loosely attached (between the rivets).		X
	Wrong seam or stitch type.	X	
	Stitching thread not black.		X
	Stitching loose, resulting in loose bobbin or top thread.		X
	Stitching tight, resulting in puckering of fabric.		X
	Open seam up to and including 1/2 inch.		X
	Open seam more than 1 inch.	X	
	One stitch per inch less than minimum specified.		X
	Two or more stitches per inch less than minimum specified.	X	
	More than two stitches per inch in excess of maximum specified.		X
	Stitching margin not as specified.		X
	Row of stitching omitted.	X	
	Stitching pattern not as specified.		X
	Thread break, not overstitched.		X
	End of stitching not backstitched.		X
	Thread ends not trimmed.		X
Rivets and washers	Rivet or washer missing.	X	
	Head of rivet on wrong side of component.		X
	Rivet not properly peened.		X
	Wrong size rivet or washer.		X

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Examine	Defect	Classification Major	Minor
Welds and welding	Not welded where required (weld missing).	X	
	Wrong type weld.		X
	Weldment not smooth or not uniform.		X
	Metal burned or damaged at weld.		X
	Weld spatter.		X
	Weld incomplete, fractured, cracked, or not fused.		X
	Weld metal in slot of pin springs.	X	
Identification marking	Missing, illegible, incomplete wrong text, not specified type or size, not applied in the specified manner, or not in location specified.	X	

4.3.4.1.2 Dimensional examination. Dimensional examination shall be made on a pair of footwear to determine compliance with this specification and drawings. Any dimension that is not within the specified tolerances shall be classified as a defect.

4.3.4.1.3 Inspection levels and AQL's. The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection Level	AQL's Major	Total
For defects applicable to 4.3.4.1.1	II	1.5	4.0
For defects applicable to 4.3.4.1.2	S-3	(one class)	4.0

4.3.5 Examination of preparation for delivery . An examination shall be made to determine that preservation, packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects shall be scored in accordance with Table VII. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed in Table VII shall be examined on shipping containers fully prepared for delivery. The lot shall be the number of shipping containers offered for delivery at one time. The inspection level shall be S-2, and the acceptable quality level (AQL) shall be 2.5 expressed in terms of defects per hundred units.

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Table VII - Examination of preservation, packing and marking

Examine	Defect
Preservation and packaging	Not as specified.
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate nailing, bulged or distorted containers.
Weight	Gross weight exceeds 200 pounds.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Preservation. Armored footwear shall be sealed in water-vapor-proof bags in accordance with method 1A8 (preservation compounds shall not be applied to armored footwear) of MIL-P-116.

5.1.1.2 Packaging. One pair of armored footwear, sealed in bags, as specified above, shall be placed within a close-fitting fiberboard box conforming to PPP-B-636, type II, class 2 or 3, grade 3 (V3s, Ref), interior box style CSSC. Fill all voids with dunnage conforming to the requirements as applicable of MIL-P-116. Dunnage shall be placed within the unit container in a manner that will provide support for all interior surfaces. Box closure shall be in accordance with the Appendix to PPP-B-636 for interior boxes. All seams and joints shall be sealed with pressure-sensitive tape conforming to PPP-T-76.

5.1.2 Level C. Each pair of armored footwear shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.2).

5.2.1 Level A. A quantity of like size armored footwear, packaged as specified in 5.1, shall be packed in close-fitting nailed wood, cleated-plywood or wood-wire-bound boxes conforming to PPP-B-621, class 2, style 4; PPP-B-601, overseas type, style A or B; or PPP-B-585, class 3. Shipping containers shall be closed and strapped in accordance with the appendix to the applicable box specification. The gross weight of shipping containers shall not exceed 200 pounds.

5.2.2 Level B. Armored footwear preserved and packaged in unit containers specified in 5.1 shall be reinforced by pressure sensitive filament tape conforming to type IV PPP-T-97 and as specified in the appendix to PPP-B-636. No further packing shall be required.

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5.2.3 Level C. Armored footwear, packaged as specified in 5.1, shall require no further packing.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The armored footwear covered by this specification is intended for protection of the lower limbs and feet against the blast effects of antipersonnel mines, or similar devices, when detonated; and is used by male personnel of the U. S. Marine Corps.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Whether preproduction sample approval is required (3.2).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 Samples. For information regarding the availability of sample footwear, address inquiry to the procuring activity issuing the invitation for bids.

6.4 Metallic components. For components such as hull braces, roll pins, etc., where limited work hardening is anticipated, a moderately work hardening alloy such as class 304, condition A of QQ-S-763 is permissible. Use of more severe work hardening alloy such as class 302, condition A of QQ-S-766 should be avoided.

Preparing activity:
Navy - MC
Project No. 8470-N001

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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